VNR Vignana Jyothi Institute of Engineering

Mechanical Engineering | Course Code: ME301

Applied Thermodynamics & Energy Conversion Cycles - Comprehensive Syllab

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Syllabus Focus: ME301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Applied Thermodynamics & Energy Conversion Cy

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: First Law of Thermodynamics & Energy Analysis

" Syllabus Scope: Thermodynamic systems, boundary work, heat interactions, First Law for closed systems,

Unit 2: Second Law, Entropy & Exergy Destruction

" Syllabus Scope: Kelvin-Planck and Clausius statements, Carnot cycle, Clausius inequality, entropy chang

ME301: Applied Thermodynamics & Energy Conversion Cycles - Advanced Topics & Ass

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Vapor Power Cycles Rankine

" Ideal Rankine cycle, superheat, reheat cycle, regenerative feed-water heating, thermal efficiency, boiler an

Unit 4: Gas Power Cycles Otto, Diesel, Dual, Brayton

" Air standard assumptions, Otto and Diesel cycles, mean effective pressure MEP, Brayton gas turbine cycle, in

Unit 5: Refrigeration & Psychrometric Principles

" Vapor Compression Refrigeration VCR, Coefficient of Performance COP, refrigerants, psychrometric chart, spec

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the thermal efficiency equation for an ideal air-standard Diesel cycle in terms of compression
2. A steam power plant operates on an ideal reheat Rankine cycle between 10 MPa and 10 kPa. Calculate net
3. Explain exergy destruction and derive the Gouy-Stodola theorem relating irreversibility to entropy gen

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Yunus A. Cengel, Michael A. Boles - Thermodynamics: An Engineering Approach McGraw-Hill 8th Ed
* P.K. Nag - Engineering Thermodynamics McGraw-Hill 5th Ed

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